

GROUIN DE CANCALE, France

WMO: 071270

Lat: 48.7098N

Lon: 1.8451W

Elev: 41

StdP: 100.83

Time Zone: 1.00 (EUC)

Period: 86-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -2.0	(c) -0.5	(d) -6.8	(e) 2.1	(f) -1.1	(g) -4.2	(h) 2.7	(i) 1.1	(j) 23.2	(k) 8.7	(l) 20.8	(m) 8.1	(n) 6.4	(o) 100	(p) 0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 4.5	(c) 23.8	(d) 19.4	(e) 22.4	(f) 18.7	(g) 21.2	(h) 18.1	(i) 20.2	(j) 22.6	(k) 19.3	(l) 21.5	(m) 18.6	(n) 20.6	(o) 4.9	(p) 40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 19.2	(b) 14.1	(c) 21.4	(d) 18.5	(e) 13.4	(f) 20.6	(g) 17.8	(h) 12.8	(i) 19.8	(j) 57.9	(k) 22.8	(l) 55.2	(m) 21.5	(n) 52.8	(o) 20.8	(p) 23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.8	(b) 16.6	(c) 14.1	(e) -4.7	(f) 27.8	(g) 3.0	(h) 2.2	(i) -6.9	(j) 29.4	(k) -8.6	(l) 30.7	(m) -10.3	(n) 31.9	(o) -12.5	(p) 33.6		
			DB													
			WB	-5.7	21.6	2.8	0.9	-7.8	22.2	-9.4	22.7	-11.0	23.2	-13.0	23.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	6.3	6.5	8.6	9.6	13.1	15.3	17.6	18.3	15.9	13.2	9.5	7.3
	DBStd	4.90	3.62	3.32	2.31	2.26	2.41	2.19	1.87	1.96	2.06	2.34	2.99	3.18
	HDD10.0	447	118	101	55	33	3	0	0	0	0	3	42	92
	HDD18.3	2433	373	331	303	263	164	95	37	24	77	159	266	342
	CDD10.0	1104	4	4	11	21	98	160	236	257	177	102	26	9
	CDD18.3	48	0	0	0	0	1	5	14	23	4	0	0	0
	CDH23.3	73	0	0	0	0	2	16	17	34	3	0	0	0
	CDH26.7	7	0	0	0	0	0	2	1	3	0	0	0	0
Wind	WSAvg	6.8	7.4	7.0	6.8	7.1	5.9	6.4	6.0	5.9	6.8	7.3	7.2	7.6
Precipitation	PrecAvg	712	61	53	44	54	52	45	45	52	61	85	80	81
	PrecMax	904	130	121	137	113	119	98	93	118	138	136	177	178
	PrecMin	552	11	10	9	17	16	15	18	14	15	20	35	22
	PrecStd	102	37	29	28	24	26	26	20	29	33	29	35	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.5	14.4	15.9	19.0	22.1	25.4	25.5	26.5	23.0	20.5	16.0	14.7
		MCWB	12.0	12.4	12.8	14.1	16.9	19.5	19.2	20.5	19.1	17.0	14.8	13.5
	2%	DB	12.7	12.8	14.1	16.0	19.6	22.2	23.2	24.0	21.3	18.5	14.7	13.6
		MCWB	11.5	11.1	12.0	12.7	15.4	17.7	19.2	19.8	18.2	16.2	13.5	12.2
	5%	DB	11.7	11.8	12.7	14.3	18.1	19.9	22.0	22.4	20.1	17.3	13.8	12.7
		MCWB	10.6	10.5	11.0	11.9	14.8	16.8	18.7	18.9	17.3	15.5	12.8	11.5
	10%	DB	10.5	10.7	11.7	12.8	16.7	18.4	20.7	21.2	19.0	16.4	13.2	11.6
		MCWB	9.4	9.6	10.1	11.0	14.1	15.9	17.9	18.3	16.7	14.9	12.1	10.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.5	12.7	13.4	14.8	17.6	20.0	20.9	21.4	20.4	17.5	15.0	13.8
		MCDB	13.2	13.9	15.2	17.4	21.1	24.2	23.0	24.8	22.2	19.0	15.6	14.4
	2%	WB	11.7	11.5	12.2	13.2	16.0	18.3	19.7	20.4	18.8	16.6	13.8	12.5
		MCDB	12.5	12.4	13.6	15.4	18.5	21.5	22.1	23.0	20.2	17.9	14.4	13.3
	5%	WB	10.7	10.6	11.3	12.2	15.1	17.2	19.0	19.6	18.0	15.8	13.0	11.6
		MCDB	11.6	11.6	12.5	13.8	17.5	19.5	21.3	21.7	19.4	16.9	13.7	12.5
	10%	WB	9.6	9.7	10.4	11.2	14.3	16.2	18.2	18.9	17.1	15.2	12.3	10.6
		MCDB	10.4	10.7	11.5	12.5	16.5	18.1	20.4	20.8	18.8	16.2	13.1	11.5
Mean Daily Temperature Range	5% DB	MDBR	3.4	3.8	4.1	4.2	4.6	4.5	4.4	4.5	4.2	3.8	3.4	3.5
		MCDBR	4.3	5.2	5.7	6.3	6.8	7.8	6.5	6.4	5.8	4.9	3.7	4.2
		MCWBR	4.0	4.3	4.1	4.2	4.1	4.3	4.0	3.5	3.5	3.5	3.4	3.9
	5% WB	MCDBR	4.1	5.0	5.3	5.5	6.1	7.0	5.8	5.6	5.1	4.5	3.6	4.1
		MCWBR	4.1	4.4	4.1	4.0	4.0	4.2	3.9	3.4	3.7	3.6	3.5	4.0
Clear-Sky Solar Irradiance	taub	0.330	0.346	0.376	0.391	0.397	0.394	0.391	0.382	0.368	0.359	0.335	0.323	
	taud	2.424	2.395	2.314	2.290	2.317	2.359	2.368	2.414	2.434	2.451	2.443	2.396	
	Ebn at Noon	727	801	832	857	864	868	865	857	834	776	713	678	
	Edh at Noon	66	85	107	121	123	119	116	106	94	79	65	60	
All-Sky Solar Radiation	RadAvg	0.96	1.73	2.84	4.36	5.16	5.71	5.47	4.66	3.66	2.10	1.17	0.80	
	RadStd	0.09	0.20	0.32	0.45	0.43	0.61	0.36	0.38	0.27	0.17	0.12	0.10	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page